

# Area of Regular Polygons

Lesson 5-9

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

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## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can find the area of a regular polygon by decomposing it into triangles.

**Language Objective:** I can explain my steps using the words regular polygon, decompose, triangle, and composite.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Area of Regular Polygons

Regular Polygon

Decompose

Triangle

Composite

Formula



Tap any word to see what it means and a picture.

1 The space inside a regular polygon, found with  $A = (\frac{1}{2}bh) \times n$ , is the

2 A polygon with all sides and all angles equal is a .

3 To break a shape into smaller shapes is to  it.

4 A three-sided polygon is a .

5 Made of two or more shapes combined, a figure is .

6 A math rule written with symbols is a .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How much will the gazebo flooring cost?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Area of Regular Polygons** — The space a regular polygon covers. Split it from the center into identical triangles, find one triangle's area, then multiply by how many there are.

*Área de polígonos regulares — El espacio que cubre un polígono regular. Divídelo desde el centro en triángulos idénticos, halla el área de uno y multiplícala por cuántos hay.*

- ☐ **Regular Polygon** — A closed flat shape made of straight sides, where every side is the same length and every angle is the same size.

*Polígono regular — Una figura plana cerrada de lados rectos, donde todos los lados miden lo mismo y todos los ángulos son iguales.*

- ☐ **Decompose** — To break a shape into smaller, simpler shapes.

*Descomponer — Separar una figura en figuras más pequeñas y simples.*

- ☐ **Triangle** — A shape with three sides.

*Triángulo — Una figura de tres lados.*

- ☐ **Composite** — Made by putting two or more simple shapes together.

*Compuesto — Formada al juntar dos o más figuras simples.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**One triangle's area is \_\_\_\_ sq ft. The octagon has \_\_\_\_ triangles, so total area = \_\_\_\_ sq ft. The cost is \_\_\_\_ x \$3 = \$ \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** We decompose the hexagon into triangles because we already know how to find a triangle's area.

**Evidence:** Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.

**Reasoning:** This evidence connects the definition of area of regular polygons to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- One triangle's area is \_\_\_\_ sq ft. The octagon has \_\_\_\_ triangles, so total area = \_\_\_\_ sq ft. The cost is \_\_\_\_ x \$3 = \$ \_\_\_\_.
- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.  
*Mi afirmación es \_\_\_\_.*
- I know because \_\_\_\_.  
*Lo sé porque \_\_\_\_.*
- So \_\_\_\_.  
*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

